



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University – Chennai • Approved by AICTE • Accredited by NAAC with "A" Grade

SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA

Ph : 04295-226000/221289 Fax : 04295-226666 E-mail : stayahead@bitsathy.ac.in Web : www.bitsathy.ac.in

Lecture Notes

22GE004 – BASICS OF ELETRONICS ENGINEERING

UNIT 3 - SIGNAL CONDITIONING USING TRANSISTOR

TRANSISTOR OPERATIONS AND LIMITATIONS

Introduction

If we read the history of electronic devices, one of the most important components of these devices was a vacuum tube (electron tube). This tube was used to control the electric current. These tubes were larger, required higher operating voltage, high power consumption meant high heat generation which in turn affected the life of the tube due to its low efficiency.

On this page, we will learn about the following:

- Transistor
- Use of transistors
- Applications of transistors

To resolve this problem, the three American physicists John Bardeen, Walter Brattain, and William Shockley invented a compact-sized and efficient semiconductor device called point-contact transistor at Bell Labs in December.

Transistor

A transistor is a kind of semiconductor that is used as a conductor and insulation of electric current or voltage. In simpler terms, a transistor is basically a regulator of the flow of electric signals. Read the following points to know more about transistor:

- Transistors are powerful devices because of their ability to control the current flowing through a circuit (current controlling device), which is generated by the flow of electrons and holes. There are two types: NPN (negative-positive-negative) and PNP (positive-negative-positive).
- The most widely used transistors are NPN transistors as the majority of charge carriers are electrons that are better mobile charge particles with less mass due to which they can easily accelerate.

- It is a semiconductor device that acts as a switch and an amplifier. Transistors can operate on a low-voltage supply for greater safety which means they yield higher efficiency and very long life.
- The transistors use semiconductor junctions instead of heating electrodes but perform the same function as a vacuum triode.
- The transistors can control the flow of current through one channel by changing the intensity of a small amount of current flowing via a second channel. That's why they are called the current controlling device.

Parts of a Transistor

A transistor is a combination of three terminals made of semiconducting materials that help in making a connection to an external circuit and allow current to flow. The three terminals are:

1. **Base:** The base activates the transistor. It is thin and lightly doped. It is put in the centre of the transistor.
2. **Emitter:** The emitter is the negative terminal of the transistor. It is heavily doped and is moderately sized.
3. **Collector:** The collector is the negative terminal of the transistor. It is located on the right side of a transistor and is moderately doped. It is larger than the emitter.

Working of a Transistor

A Bipolar Junction Transistor or BJT consists of three terminals- base, emitter and collector. A p-n junction exists between base and emitter, and another junction exists between base and collector. Normally, in BJT when current flows through the base-emitter junction, a current will flow in the collector circuit. This is called bias and the base-emitter junction is forward biased whereas the base-collector junction is reverse biased.

Basics of Bipolar Junction Transistors

Since the controlled current must go through two types of semiconductors materials the current consists of both electron and hole flow, in different parts of the transistor, and these are of two types:

1. n-p-n junction transistor
 2. p-n-p junction transistor
- The base is responsible for activating the transistor.
 - The emitter is the negative lead while the collector is the positive lead.

Isolated Gate Bipolar Transistor (IGBT): IGBT is a power semiconductor device used as an electronics switch in much high power and modern appliances such as electric cars, trains, variable speed refrigerators, air conditioning systems.

What are the Characteristics of a Transistor?

The characteristics of a transistor is the graph plotted for each type of configuration, which shows the relationship between the current and voltage of the transistor.

There are mainly two types of characteristics:

- I. **Input characteristics:** This shows the change in input current with varying output current when the output voltage is constant.
- II. **Output characteristics:** This graph shows a plot of changing output current with respect to a change in output voltage when the input current is constant.

Advantages of using Transistors

Transistor has been proven as a very important invention in science. It has many uses and advantages:

- It is small in size and is very cost-efficient.
- It needs very low voltage to function.
- It has a long life and requires no power to operate.
- A single integrated circuit can be developed using the transistor.
- Current switches fast in the terminals.

Limitations of using Transistors

Even though transistors are extremely efficient, there are some limitations to its uses:

- Transistors get damaged very easily due to changes in electrical and temperature conditions.
- They lack higher electron mobility.
- They can get affected by radiation.

Know more about transistors by visiting our website where you can find notes, questions, answers and solutions and more! You can download anything you need for free!

Uses of Transistor

- **As Amplifier Circuit:** A transistor can be used to amplify current.

V_{be} biasing voltage produced in the base-emitter junction. Due to the forward biasing of the base-emitter junction, the electrons start flowing from emitter to recombine with holes in the base, the base becomes negatively charged. If the base current I_b is increased by a small amount, hole-electron recombination will get neutralised, the collector current I_c will be increased. Therefore, a small change in current I_b in the base.

- **Microphone:** The microphone is a transducer that converts our voice or sound wave to an electronic signal. As the sound wave doesn't have a constant value, the magnitude of the sound wave varies with time according to our voice.

The electrical output of the microphone varies according to the sound waves as the base current I_b is varying because of the small alternating voltage produced by the microphone which means a small change in I_b can cause a large change in I_c .

When this output of the microphone is given to the transistor as an input. The varying collector current I_c flows into the loudspeaker, and we know that if there are changes in the input of the transistor there will be a large change in the output of the transistor. Thus, the transistor amplifies the electronic signal of the microphone.

The frequency remains constant but the amplitude of the sound wave from the loudspeaker is higher than sound waves fed into the microphone.

- **Oscillator Circuit**

Figure.2 shows a schematic representation of an oscillating circuit.

An electronic oscillator is a device that generates continuous electrical oscillations. In a simple oscillator circuit, a parallel LC circuit is used as a resonant circuit and an amplifier is used to feed energy to the resonant circuit.

The frequency gets resonantly amplified, and the output acts as a source of an alternating voltage of that frequency.

- **Transistor Used as a Switch**

BJT Transistors can be used as a switching device to control DC power to a load. The switched (controlled) current goes between emitter and collector, and the controlling current goes between emitter and base.

- In normal operation of a transistor, the emitter-base junction is always forward-biased whereas the collector-base junction is reverse biased.
- In n-p-n junction transistors, there are a large number of electrons in the emitter and a large number of holes in the base.
- In the actual design of n-p-n transistors, the middle layer is very thin (micrometre) as compared to the widths of the two layers at the sides.

ADVANTAGES AND DISADVANTAGES OF TRANSISTORS

Transistors are devices that act as switches or gates to control the flow of electrical voltage and current. They're made from semi conductive materials and are essentially miniature semiconductors, each featuring at least three terminals connecting to an external circuit. Characteristically consisting of three semi conductive layers, transistors act as amplifiers or switches capable of carrying electrical currents. Transistors convert smaller input currents into larger output currents when working as amplifiers. If operating as switches, they exist in one of two distinct states – either on or off – that control the flow of electricity through electronic devices or electrical circuits.

There is only one circuit element on each transistor, and a small number of transistors can be combined to act as electronic switches within integrated circuits. These integrated circuits are made up of numerous interconnected transistors that are then baked into silicon to create different types of microprocessors. Millions of these transistors exist on the computer chips that store memory and control processes for a wide range of modern electronic devices like smartphones, gaming consoles, flash drives, and digital cameras. They convert alternating current into direct current in power supply inverters and other high-

power and low-frequency applications. They also generate radio waves in oscillator circuits and other high-frequency applications.

Advantages of Transistors

Transistors' versatility has made them an integral element of modern electronics and all computing technology. Now, let's look at some of the advantages and disadvantages of transistors...

Some advantages of using a transistor include:

- Allows for easy testing with a multimeter.
- Can be used to regulate power delivery to the electrical load in choppers and inverters.
- Can quickly switch between the “on” state to carry large currents while conversely blocking very high voltages when in the “off” state.
- Capable of operating at switching frequencies between about 10 to 15 KHz.
- The compact size means putting tens of thousands (or more) of transistors on a circuit board to create an electrical circuit is possible.
- Decades of use have resulted in the development of many types of transistors used in countless applications.
- Exceptional capabilities for switching frequencies make them ideal switches.
- Inexpensive, quick, and soundless means for controlling high-power circuitry.
- Low operational voltage makes it safe and suitable to fit into tight spaces.
- Requires minimal supply of voltage while providing high voltage gain.
- Serves a wide variety of power applications, from just a handful to several hundred kilowatts.
- Significantly longer lifespans than vacuum tubes, which they replaced, with some devices with transistors still operational after 30 years.
- Simple and durable design with no moving parts allows them to withstand vibrations, minor collisions, and other wear-and-tear types while requiring very little maintenance.
- Unlike older, tube-based radios and televisions, transistors in these applications require no cathode heater.
- Very low drops in voltage, or output impedance, when in the “on” state.
- Its very small size makes it perfect for various applications, including analog and digital circuits, cell phones, computers, microprocessors, power regulators, radar, and radios.
- Works well in cold temperatures, so doesn't require a heating element, while output voltage remains constant when temperatures rise or fall.

Essentially transistors do the same thing as two [P-N diode junctions](#) that are stacked next to each other but in opposite directions. Created via a method known as “doping,” a transistor's P-N junction marks the

boundary between two different semiconducting materials within a semiconductor. Transistors are considered either NPN or PNP, with PNP transistors having a lower current-to-voltage rating than NPN transistors, which tend to be more desirable for power conversion applications. PNP transistors switch on with a low signal. In contrast, NPN transistors switch on via a high signal, with the P representing positive polarity in the emitter terminal and N representing the negatively charged coating at the base terminal.

Transistors are also primarily made from silicon, a common element that resists the conduction of electrical current. Silicon allows transistors to operate at higher junction temperatures and currents, making them useful for alternating current applications, which require ranges up to several hundred kilowatts.

Disadvantages of Transistors

While transistors offer many advantages, the disadvantages of transistors should be contemplated when considering their use in an application.

Some disadvantages of using transistors include:

- As they produce very little energy, semiconductor fuses cannot protect transistors.
- The current capacity in [reverse blocking mode](#) is very low.
- DC output voltage isn't adjustable, so their output voltage can't easily be changed.
- Difficult to identify and repair faulty capacitors due to their small size.
- Don't operate well over a switching frequency of about 15 kHz.
- Low input impedance except for the field-effect transistors often used for digital integrated circuits.
- Must be protected against high voltages and currents, which can damage transistors.
- Sensitive to radiation and voltage stress while also being prone to secondary breakdowns due to thermal runaway.
- Techniques for manufacturing capacitors require automation and "clean room" environments, along with stringent quality control.
- Very strong electrical currents can cause transistors to heat up and break down.
- The higher the frequency when switching, the greater the loss of power.
- When switched "on," there is always voltage crossing the transistor. When switched "off," there's always a slight leakage of current.
- Comparatively slow switching time compared to the high alternating frequency of current and voltage.

Most transistor failures occur due to excess dissipation, either directly or indirectly, or because applied voltages exceed a transistor's maximum design limitations. For example, most transistors cannot block reverse voltages over 20 V, which can easily damage the device and transistors. For this reason, transistors shouldn't be used for control applications when working with alternating current unless

utilizing a reverse shunting diode to connect between the collector and emitter. Despite these difficulties when using transistors in certain applications, they've proved invaluable and have had an outsized impact on advancing modern electronics and electrical technology.

Thermal runaway in a transistor

Thermal runaway is a simple process of feedback. The part gets hot, it dissipates more power, gets hotter, runs away. For example take a reversed bias diode. The leakage current will create heat. The heat increases the leakage.